Question:

An automobile battery produces a potential difference (or "voltage") of 12 volts between its terminals. (It really consists of six 2 volt batteries following one after the other.) A headlightbulb is to be connected directly across the terminals of the battery and dissipate 40 watts of joule heat. What current will it draw and what must its resistance be?

Answer Choices:

A. 3.0 amps, 4 ohms

B. 2.8 amps, 3.2 ohms

C. 3.75 amps, 3.2 ohms

D. 4 amps, 3 ohms

E. 3.33 amps, 3.6 ohms

F. 5 amps, 2.4 ohms

G. 4.5 amps, 2.67 ohms

H. 2.2 amps, 5.45 ohms

I. 2.5 amps, 4.8 ohms

J. 3.5 amps, 4 ohms

Reasoning + Answer:

<think>

Use OhmâĢĻs law for a direct current circuit: the current through the bulb equals the batteryâĢĻs terminal voltage divided by the bulbâĢĻs resistance.

- Power across the terminal pair is P = V ÃĹ I.
- Here, P = 40 W and V = 12 V, so I = P/V = 40/12 = 10/3 âīĪ 3.33 A.
- Then the bulbâĢĻs resistance is R = V/I = 12/(10/3) = 3.6 Î©.

Therefore, the correct answer is E. 3.33 amps, 3.6 ohms.

</think>

<answer>

E. 3.33 amps, 3.6 ohms

</answer>